CASE STUDY REPORT

Website & Application Feature Analysis

Case Study 1

Uber – Ride Fare Estimation System

Feature Focus: Dynamic Pricing Algorithm

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Class: CT-B | Roll Number: 178

Project Type: Website & Front-End Application

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1. Introduction

Uber is a web-based ride-hailing platform that allows users to book rides by entering pickup and drop locations. The system calculates fares in real time and provides multiple ride options. This case study focuses on the Dynamic Pricing Algorithm used in the Uber website for fare estimation.

2. Problem Statement

In real-world transportation systems, demand for rides varies depending on time, location, weather, and events. Fixed pricing fails to balance demand and supply, leading to unavailability of drivers or long waiting times. A system is therefore required that dynamically adjusts prices to ensure efficient ride allocation.

3. Objectives

- To analyze how Uber calculates ride fares dynamically.
- To understand the role of demand and supply in pricing.
- To study how algorithms are reflected in the website UI.

4. Proposed Solution

Uber uses a Dynamic Pricing Algorithm (also known as Surge Pricing) to adjust fares based on real-time conditions. The algorithm increases prices when demand is high and decreases them when demand is low, ensuring availability of drivers and reduced waiting time.

5. Feature Description – Dynamic Pricing

When a user enters pickup and drop locations on the Uber website, the system performs the following:

- Calculates distance and estimated travel time.
- Checks nearby driver availability.
- Analyzes current demand in the area.
- Generates a fare for each available ride option based on these factors.

6. Algorithm Logic

Core decision rule:

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IF (Number of Ride Requests > Available Drivers) → Increase Price
(Surge)
ELSE → Keep Price Normal
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Additional factors considered by the algorithm:

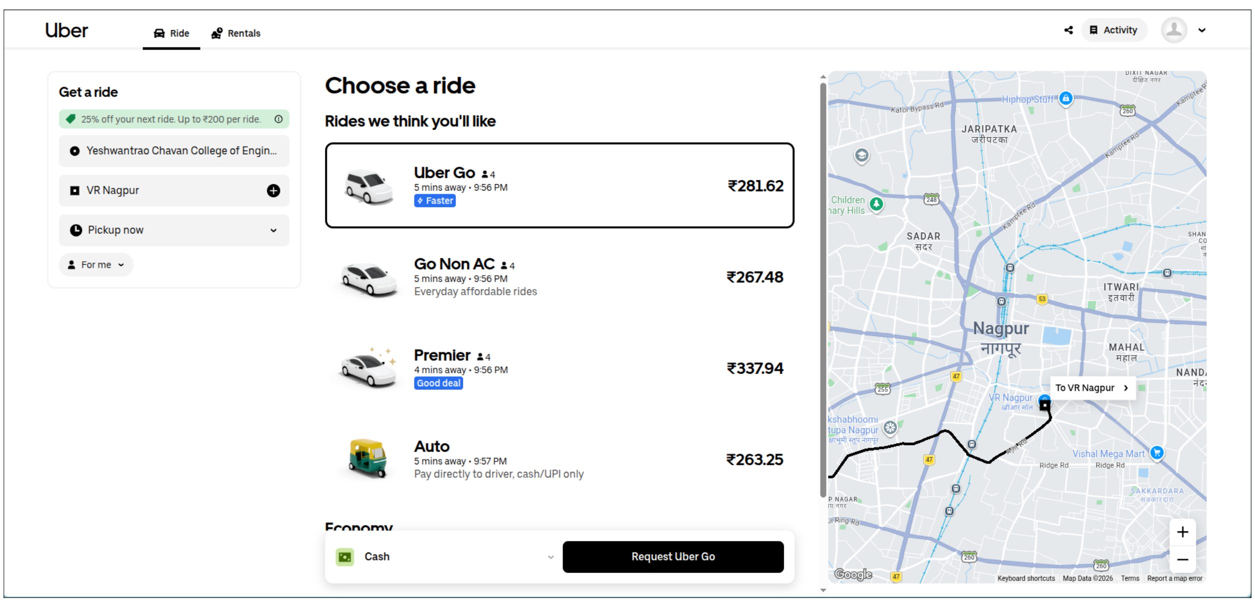
- Distance between pickup and drop location.
- Time duration of the trip.
- Traffic conditions along the route.
- Ride type selected (Auto, Go, Premier, etc.).

7. UI Evidence (From Website)

Observations from the Uber website interface:

- Multiple ride options display different prices simultaneously.
- Prices change dynamically based on current conditions.
- Estimated arrival times vary by ride type.

This demonstrates that pricing is not fixed and is entirely controlled by the underlying algorithm. The screenshot below (from the Uber website, showing a route to VR Nagpur) captures multiple ride options with varying fares — Uber Go at ₹281.62, Go Non AC at ₹267.48, Premier at ₹337.94, and Auto at ₹263.25 — confirming dynamic pricing in action.



8. Working of the System

Step-by-step flow of the Uber dynamic pricing system:

1. User enters pickup and drop location on the Uber website.
2. System calculates the route using a shortest path algorithm.
3. System checks current demand and driver supply in the area.
4. Dynamic pricing algorithm calculates the appropriate fare.
5. Website displays multiple ride options with their computed prices.

9. Advantages & Limitations

Advantages	Limitations
Efficient resource utilization	Prices can become very high during peak times
Reduced waiting time for users	Users may find pricing unpredictable
Fair distribution of rides	Surge pricing can feel unfair during emergencies
Real-time responsiveness to demand	Limited control for users over price fluctuations

10. Conclusion

Uber's dynamic pricing algorithm plays a crucial role in balancing demand and supply in real time. It ensures availability of rides while optimizing pricing conditions. The effect of this algorithm is clearly visible in the website UI through changing fares and ride options presented to users at the time of booking.